
Exercise 0

1 Understanding MPC

1. What are the advantages and disadvantages of using MPC instead of using classical controllers such as PID?
2. Explain how the receding-horizon works, what are the advantages and disadvantages of choosing a long horizon?
3. What is the difference between open-loop optimal control and MPC? Why does MPC recompute the control sequence at each time step instead of using the whole open-loop plan?

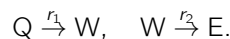
2 State space model formulation

In this lecture, we use the state space representation of a discrete dynamic system. For a linear time-invariant system (LTI system), it has the form

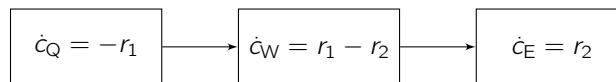
$$\begin{aligned}x_{k+1} &= Ax_k + Bu_k, \\y_k &= Cx_k + Du_k,\end{aligned}$$

where $x \in \mathbb{R}^{n_x}$ is the system state, $u \in \mathbb{R}^{n_u}$ is the system input, $y \in \mathbb{R}^{n_y}$ is the output. This exercise will demonstrate how to derive such a model for a simple chemical process.

Consider the following chemical reaction kinetics for a two-step series reaction with three species Q, W and E



The material balances for the three species are



We could discretize these three ordinary differential equations and obtain the discrete system

$$\frac{c_Q(k+1) - c_Q(k)}{\Delta t} = -r_1, \quad \frac{c_W(k+1) - c_W(k)}{\Delta t} = r_1 - r_2, \quad \frac{c_E(k+1) - c_E(k)}{\Delta t} = r_2,$$

in which c_j is the concentration of species j , and r_1 and r_2 are the rates (mol/(time · vol)) at which the two reactions occur, Δt is the discretization time interval, which is a known constant. We assume the rate law for the reaction kinetics is

$$r_1 = c_Q(k), \quad r_2 = c_W(k).$$

Write the linear state space model for this chemical reaction model. Assume we can measure c_Q , the concentration of component Q. What are x , y , A , and C for this model? (Note that this system does not have inputs.) (Adapted from Exercise 1.1 in Model Predictive Control: Theory, Computation, and Design by J. B. Rawlings, D. Q. Mayne, and M. Diehl)